

Diagram illustrating the PMOS network (2T1C1) for the function $f(a, b) = a + b$. The network consists of two PMOS transistors, H1 and H2, connected in parallel between VDD and GND. H1 has a minus sign (-) on its gate, and H2 has a plus sign (+) on its gate. A capacitor C3 is connected between the output node and GND, labeled C_Polarized.

Diagram illustrating the connection of a USB-to-serial adapter module (J1) to a USB port (D5).

The module (J1) is connected to the USB port (D5) as follows:

- VBUS:** Connected to pin A4 of the USB port. A diode is connected in series, with the cathode to the module and the anode to the 5V+ supply.
- CC1:** Connected to pin A5 of the USB port.
- CC2:** Connected to pin B5 of the USB port.
- D-:** Connected to pin A7 of the USB port.
- D+:** Connected to pin B7 of the USB port.
- SBU1:** Connected to pin A8 of the USB port.
- SBU2:** Connected to pin B8 of the USB port.
- Shield:** Connected to pin A1 of the USB port.
- Ground:** Connected to the ground pin of the USB port.

The module (J1) is labeled with "SHIELD" and "A1". The USB port (D5) is labeled with "5V+", "A4", "A5", "B5", "A7", "B7", "A6", "B6", "A8", and "B8". The module also has a USB symbol and a shield symbol.

Diagram illustrating the USB interface circuit for the FT232RL module. The circuit shows the connection of the USB pins (D-, D+, GND, VCC, RESET) to the FT232RL chip (U9).

Components:

- Resistors: R12 (27Ω), R13 (27Ω)
- Capacitors: C7 (47pF), C8 (47pF)

Pin Connections:

Module Pin	Signal	Chip Pin	Notes
20	RX0	TXD	
4	TX0	RXD	
2	RTS	RTS	
9	DTR	CTS	
7	DSR	DTR	✗
5	DCD	DSR	✗
18	CBUS0	DCD	✗
17	CBUS1	RI	✗
10	CBUS2		✗
19	CBUS3		✗

Power and Ground Connections:

- VCC:** 3V3OUT (Pin 13) connected to +3V3.
- RESET:** Pin 14 connected to RESET.
- GND:** Pins 6 and 16 connected to GND.

USB Signal Connections:

- D-:** Connected to Pin 12 via resistor R12.
- D+:** Connected to Pin 11 via resistor R13.
- GND:** Connected to Pins 13 and 14 via capacitors C7 and C8.

3 state buffer

U7 TC7WH241FK +3V3

EN1 1 G VCC 8 EN1
ICS1_3V3 2 A1 G 7 EN2
ICS1_3V3 3 Y1 Y1 6 RX1
4 GND A2 5 TX1

GND

U8 TC7WH241FK +3V3

EN2 1 G VCC 8
ICS2_3V3 2 A1 G 7 EN2
ICS2_3V3 3 Y2 Y1 6 RX2
4 GND A2 5 TX2

GND

The diagram shows an ESP32-S3-WROOM-1 module with the following connections:

- Power:**
 - VCC (EN 3):** Connected to +3V3.
 - GND (EN 2):** Connected to GND.
 - BOOT (27):** Connected to GND.
 - BNO_RST (7):** Connected to +3V3.
- GPIOs:**
 - IO0 (100):** Connected to RXD0 (37).
 - IO1 (101):** Connected to RXD0 (36).
 - IO2 (102):** Connected to TXD0 (101).
 - IO3 (103):** Connected to TXD0 (11).
 - IO4 (104):** Connected to TXD1 (14).
 - IO5 (105):** Connected to TXD1 (13).
 - IO6 (106):** Connected to TXD1 (12).
 - IO7 (107):** Connected to TXD1 (11).
 - IO8 (108):** Connected to TXD1 (10).
 - IO9 (109):** Connected to TXD1 (9).
 - IO10 (110):** Connected to TXD1 (8).
 - IO11 (111):** Connected to TXD1 (7).
 - IO12 (112):** Connected to TXD1 (6).
 - IO13 (113):** Connected to TXD1 (5).
 - IO14 (114):** Connected to TXD1 (4).
 - IO15 (115):** Connected to TXD1 (3).
 - IO16 (116):** Connected to TXD1 (2).
- Other Connections:**
 - USB_D- (103):** Connected to USB_D+ (104).
 - PSRAM (103):** Connected to PSRAM (104).
 - SW (103):** Connected to SW (104).
 - EN_Power (103):** Connected to EN_Power (104).
 - led (103):** Connected to led (104).
 - RGB_led (103):** Connected to RGB_led (104).

The diagram shows the DM3AT-SF-PEJM5 module with the following connections:

- Pin 11:** SWB, connected to +3V3 through a D8 LED and resistor R18 (10k), and to GND through switch SWA.
- Pin 10:** SWB, connected to GND.
- Pin 9:** SWA, connected to GND.
- Pin 3:** MOSI, connected to GND.
- Pin 2:** CS, connected to GND.
- Pin 1:** DAT2, connected to GND.
- Pin 8:** DAT1, connected to GND.
- Pin 7:** DAT0, connected to GND.
- Pin 5:** CLK, connected to GND.
- Pin 4:** VDD, connected to +3V3.
- Pin 6:** VSS, connected to GND.



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